

FGeneBERT: function-driven pre-trained gene language model for metagenomics

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Abstract

Metagenomic data, comprising mixed multi-species genomes, are prevalent in diverse environments like oceans and soils, significantly impacting human health and ecological functions. However, current research relies on K-mer, which limits the capture of structurally and functionally relevant gene contexts. Moreover, these approaches struggle with encoding biologically meaningful genes and fail to address the one-to-many and many-to-one relationships inherent in metagenomic data. To overcome these challenges, we introduce FGeneBERT, a novel metagenomic pre-trained model that employs a protein-based gene representation as a context-aware and structure-relevant tokenizer. FGeneBERT incorporates masked gene modeling to enhance the understanding of inter-gene contextual relationships and triplet enhanced metagenomic contrastive learning to elucidate gene sequence–function relationships. Pre-trained on over 100 million metagenomic sequences, FGeneBERT demonstrates superior performance on metagenomic datasets at four levels, spanning gene, functional, bacterial, and environmental levels and ranging from 1 to 213 k input sequences. Case studies of ATP synthase and gene operons highlight FGeneBERT’s capability for functional recognition and its biological relevance in metagenomic research.

Keywords: metagenomics; DNA; pre-trained language model; transformer

Introduction

Metagenomics, the study of mixed genomes of microbial communities in the environment (e.g. gut microbiomes or soil ecosystems) [1–3], has revealed the critical role in fundamental biological processes like enzyme synthesis, gene expression regulation, and immune function [4–6]. This deepened understanding highlights the need to accurately interpret the intricate genetic information contained within these diverse communities. Consequently, deciphering the complex sequences of multiple species in metagenomics is vital for unraveling life’s mechanisms [7, 8], and advancing biotechnology [9, 10], with important applications in biodiversity conservation and epidemiology [11].

Unlike traditional genomics focused on single species, metagenomics involves genetic material directly from environmental samples, posing significant challenges due to sample diversity and species abundance [12, 13]. As shown in Fig. 1, the typical challenges in metagenomics are the presence of one-to-many (OTM) and many-to-one (MTO) problems. The OTM problem indicates that a single gene can exhibit various functions in different genomic contexts, underscoring the significance of

inter-gene interactions in function regulation [14]. For example, ATP synthase displays distinct functionalities in diverse organisms such as bacteria, plants, and humans (Fig. 1a). Conversely, the MTO problem implies that different genes can share the same function, emphasizing expression commonality [15]. For example, the CRISPR (CRISPR-Cas systems are essential adaptive immune components in microorganisms that defend against mobile genetic elements and form the basis of advanced genome engineering technologies. Unlike the commonly used DNA-targeting Cas9 or Cas12 systems, Cas13 is an RNA-guided, programmable RNA-targeting system that enables gene manipulation at the transcriptional level.) immune mechanism [15, 16] involves various proteins like Cpf1, Cas1, and Cas13, each contributing to the same defensive function (Fig. 1b) [17, 18].

Recently, various computational methods have emerged for genomic and metagenomic data analysis. Traditional alignment-based methods like MetaPhlAn4 [19] aim to match similarities between query sequences and known reference genomes and are common for taxonomic profiling. Advancements in deep learning have led to new methods like CNN-MGP [20] and DeepVirFinder

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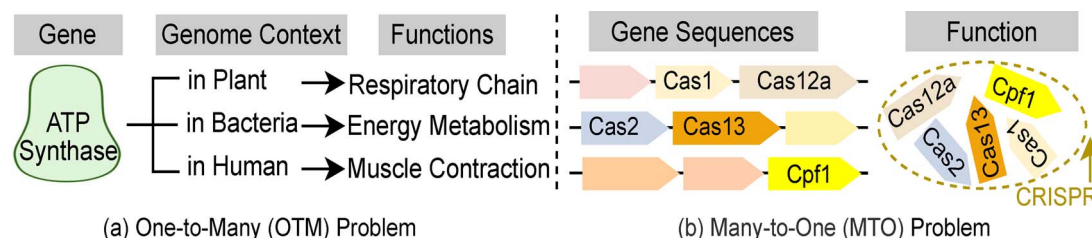


Figure 1. Motivation. Two types of complex relationships between gene sequences and functions in metagenomics. **One-to-Many** problem means that the same gene may display different functions based on the genomic context; for example, ATP synthase works differently in plants, heterotrophic bacteria, and humans. **Many-to-One** problem shows that multiple genes may perform the same function; for instance, different genes from different bacteria, e.g., Cpf1, Cas1, etc., produce the same resistance function within the immune system CRISPR.

[21], which use CNNs for gene and viral classifications with one-hot encoding. K-mer tokenization [22], employed in approaches like MDL4Microbiome [23], is a standard for DNA sequence characterization. Additionally, Virtifier [24] maps a nucleotide sequence using a codon dictionary combined with LSTM to predict viral genes. DeepMicrobes [25] employs a self-attention mechanism, while DeepTE [26] uses K-mer inputs with CNNs for element classification, and Genomic-nlp [27] applies word2vec for gene function analysis. MetaTransformer [28] uses K-mer embedding for species classification with Transformer. For pre-training models, LookingGlass [29] uses a three-layer LSTM model for functional prediction in short DNA reads. ViBE [30] employs a K-mer token-based BERT model pre-trained with Masked Language Modeling for virus identification. The BERT model, effective in DNA sequence characterization, is limited by the Transformer architecture's computational burden. LOGO [31] addresses this by cutting off long sequences into 1-2kb sub-sequences. Enformer [32] combines extended convolution with Transformers for long human genomic data. GenSLMs [33] introduce hierarchical language models for whole-genome modeling. DNABERT [34], the first pre-trained model on the human genome that focuses on extracting efficient genomic representations. DNABERT2 [35], its successor, uses Byte Pair Encoding on multi-species genomic data. NT [36] is trained on nucleotide sequences from humans and other species and evaluated on 18 genome prediction tasks. HyenaDNA [37] presents a long-range genomic model based on single-nucleotide polymorphisms on human reference genomes.

Despite these advancements, current approaches exhibit three critical limitations when analyzing metagenomic data. Firstly, the Semantic Tokenizer problem. Most machine learning-based methods [25, 27, 29], taking K-mer counts, frequencies, or embeddings as input features, providing efficient alternatives to traditional homology searches against reference genome databases. However, K-mer features often have limited representation ability and fail to capture global information. Secondly, the function-driven modeling problem. Although recent Transformer models excel in modeling complex DNA contexts through long-range dependencies, they are predominantly tailored for single-species genomic analysis [35, 36, 38]. Some models have also been developed for long-range DNA-related tasks, demonstrating promising performance in long-range species classification tasks [39, 40], although the practical applications of this problem remain poorly defined [41]. However, these tasks do not adequately address OTM and MTO challenges. This limitation impedes their ability to accurately model the intricate relationships between genes, their function across diverse genomic environments, and their connections among sequences with similar functions. Thirdly, the low generalization problem persists in metagenomic models, with recent work [42] proposing optimized strategies to enhance

model adaptability. Models like MetaTransformer [28] and ViBE [30], designed for specific tasks such as read classification and virus category prediction, fail to grasp the broader biological complexities of metagenomic data, limiting their generalization across diverse metagenomic tasks.

To address these challenges, we propose FGeneBERT (Function-Driven Pre-trained Gene Language Model for Metagenomics), a novel metagenomic pre-trained model designed to encode contextually aware and functionally relevant representations of metagenomic sequences. First, to solve the problem of encoding gene sequences with biological meaning, we propose a protein-based gene representation as a context-aware tokenizer, allowing for a flexible token vocabulary for longer metagenomic sequences. This strategy leverages the inherent protein functional and structural information encoded in metagenomic data [43], overcoming the limitations of K-mer methods and maintaining functional consistency despite potential mutations [44]. Second, we propose two pre-training tasks for function-driven modeling: masked gene modeling (MGM) and triplet enhanced metagenomic contrastive learning (TMC) to enhance the co-representation learning of metagenomic gene sequences and functions. Thirdly, FGeneBERT is pre-trained on over 100 million sequences, showcasing robust performance across diverse datasets spanning gene, functional, bacterial, and environmental levels.

In this work, we identify three key challenges in metagenomic analysis. To address these issues, we propose FGeneBERT. To the best of our knowledge, this is the first metagenomic pre-trained model encoding context-aware and function-relevant representations of metagenomic sequences. To summarize: (i) We introduce a new idea of protein-based gene representations to learn biologically meaningful tokenization of long sequences. (ii) We propose MGM to model inter-gene relationships and TMC to learn complex relationships between gene sequences and functions. (iii) We conduct extensive experiments across various downstream tasks, spanning gene, functional, bacterial, and environmental levels with input sizes from 1 to 213 k sequences. FGeneBERT achieves the state-of-the-art performance.

Methods

In this section, we provide a detailed description of the proposed pre-training model FGeneBERT, which contains the MGM and TMC components as depicted in Fig. 2.

Datasets

We pre-train and evaluate FGeneBERT on distinct datasets ranging from thousands to hundreds of thousands of sequences. For pre-training, we utilize the MGnify database [updated February 2023 (https://ftp.ebi.ac.uk/pub/databases/metagenomics/peptide_database/2023_02/)], comprising 2 973 257 435 protein

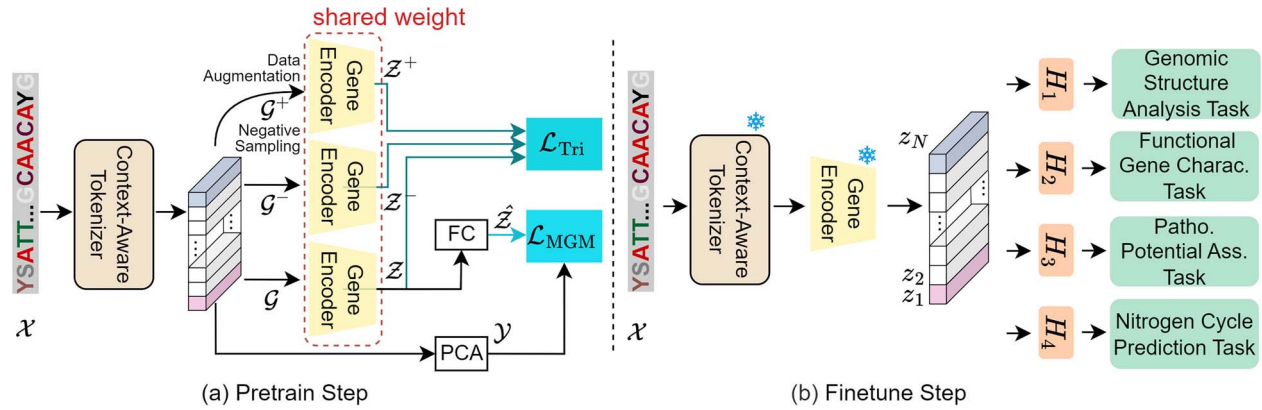


Figure 2. Overview of FGeneBERT. A metagenomic sequence $\mathcal{X}$ is converted into ordered protein-based gene representations $\mathcal{G}$ via a Context-Aware Tokenizer. Next, we pre-train a Gene Encoder with $\mathcal{L}_{\text{MGM}}$, 15% of these tokens are masked to predict labels $\mathcal{Y}$. Meanwhile, we introduce $\mathcal{L}_{\text{Tri}}$ to distinguish gene sequences. The data augmentation and negative sampling modules generate positive samples $\mathcal{G}^+$ and negative samples $\mathcal{G}^-$, respectively. Finally, after fine-tuning, FGeneBERT can handle various downstream tasks.

Table 1. Microbial species and genome numbers in each environment in the MGnify dataset. Columns denote environmental catalogs, microbial species, and assembled genomes, respectively

Catalogues	Species	Genomes
Human gut	4744	289 232
Human oral	452	1225
Cow rumen	2729	5578
Marine	1496	1504
Pig gut	1376	3972
Zebrafish fecal	79	101
Non-model fish gut	172	196

sequences aggregated from 30 FASTA files of diverse microbial communities. Table 1 details microbial genomes across environments (e.g. Human gut: 289 232 genomes, 4744 species; marine: 1504 genomes, 1496 species), distinct from its protein sequences used for pre-training, with cataloged microbial species provided. For downstream evaluation, we test FGeneBERT across four task levels—(i) gene structure analysis, (ii) functional gene prediction, (iii) pathogenicity potential assessment, and (iv) nitrogen cycle prediction—using datasets split into training, validation, and test sets (8:1:1) with five-fold cross-validation. As detailed in Table 2: E-K12 [45] (4315 operons) for gene structure; CARD [46] (1966 sequences) for functional prediction across 269 AMR families (CARD-A), 37 drug classes (CARD-D), and 7 resistance mechanisms (CARD-R); VFDB core [47] (8945 sequences, 15 VF categories) for virulence factors; ENZYME [48] (5761 sequences, 7 categories) for enzyme functions with unique EC numbers [49]; PATRIC core [50] (5000 sequences, 110 classes) for pathogenicity; and NCycDB [51] (213 501 sequences, 68 gene families) for nitrogen cycle processes (see Supplementary section *Overview of Downstream Tasks*).

Notation

Given a dataset of metagenomic long sequences $\{\mathcal{X}_i\}_{i=1}^m$, we simplify each $\mathcal{X}_i$ into a set of shorter gene fragments $\{x_i\}_{i=1}^{n_i}$ through ultrasonic-based DNA fragmentation and subsequent assembly procedures [52], where n_i represents the variable number of gene sequences derived from each $\mathcal{X}_i$. Each gene sequence x_i is tokenized in a vector $g_i \in \mathbb{R}^d$, where d is the token dimension. Each sequence is associated with a reduced-dimensional representation $y_i \in \mathbb{R}^{100}$. Suppose a gene group $\mathcal{G} = \{g_i\}_{i=1}^N$ is formed by concatenating N gene vectors sequentially. During the

pre-training phase, each gene token $g_i \in \mathcal{G}$ is processed by the context-aware genome language encoder $\mathcal{F}(\cdot)$, generating the knowledge representations z_i , where $z_i = \mathcal{F}(g_i)$. These representations are then transformed by a fully connected layer into $\hat{z}_i$, defined as $\hat{z}_i = \mathcal{H}(z_i)$, where $\mathcal{H}(\cdot)$ represents a multi-classification head. In addition, we incorporate contrastive learning into the methodology. For each gene x_i , we introduce a data augmentation module to generate positive samples $x_{j(i)}$ and a hard negative sampling strategy for constructing negative samples $x_{k(i)}$.

Context-aware masked gene modeling for one-to-many problem

Context-aware tokenizer

To develop a biologically meaningful tokenization of long sequences, we design a context-aware tokenizer utilizing the protein language model (PLM), such as ESM-2 [53] with 15 billion parameters, integrating biological prior knowledge. As illustrated in Fig. 3, this tokenizer framework begins by extracting DNA gene sequences $\{x_i\}_{i=1}^{n_i}$ from a metagenomic sequence $\mathcal{X}_i$ sourced from MGnify, which archives and preprocesses data in collaboration with the European Nucleotide Archive (ENA) software [54]. This conversion enhances the flexibility of analyzing longer metagenomic sequences.

Secondly, these DNA sequences x_i are translated into amino acid (AA) sequences using the Transeq software [55]. This translation addresses the issue of degenerate DNA codes, where certain non-“ATCG” symbols like “Y” or “S” can represent multiple nucleotides (e.g. “Y” can be “C” or “T”). By translating to AA sequences, we eliminate this redundancy, as different DNA sequences can map to the same AA sequence, ensuring consistency in representing biologically equivalent sequences [56, 57]. Thirdly, these AA sequences are transformed into 1280D normalized ESM-2 representations, with an additional gene orientation vector, resulting in 1281D gene representations $\{g_i\}_{i=1}^{n_i}$. The utility of ESM-2 lies in its ability to reveal gene relationships and functional information inherent in metagenomic data [43], preserving important intra-gene contextually-aware information. Finally, every N representations are concatenated sequentially to form gene groups $\mathcal{G}$, which serve as the basis for subsequent modeling tasks.

Masked gene modeling

We propose the MGM to enhance the model’s understanding of the relationships between genes within metagenomic sequences

Table 2. Description of experimental datasets. #Seq. and #Class means the number of sequences and categories in each dataset

Task	Dataset	Description	#Seq.	#Class
Gene structure prediction (Gene level)	E-K12	Gene operons	4315	1379
Functional prediction (Functional level)	CARD-A	AMR family	1966	269
	CARD-D	Drug class	1966	37
	CARD-R	Resistance mech.	1966	7
	VFDB	Virulence fact.	8945	15
	ENZYME	Enzyme func.	5761	7
Pathogenicity prediction (Bacteria level)	PATRIC	Pathogenic genes	5000	110
Nitrogen cycle prediction (Environmental level)	NCycDB	Cycling genes	213 501	68

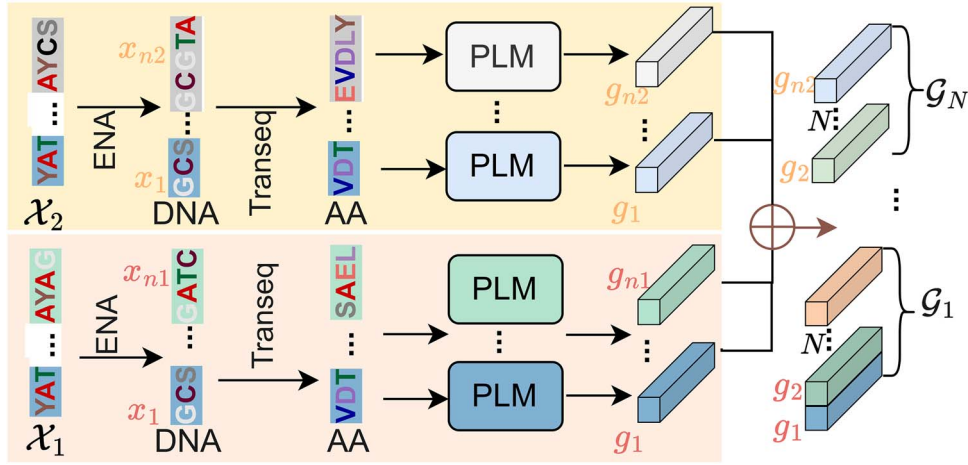


Figure 3. Framework of Context-Aware Tokenizer. Gene sequences $\{x_i\}_{i=1}^{n_i}$ extracted from metagenomic sequence $\mathcal{X}_i$ are translated into amino acid sequences and encoded into ESM-2 representations as $\{g_i\}_{i=1}^{n_i}$. These representations are then concatenated sequentially every N representation to form gene groups $\mathcal{G}$.

and their function regulations across diverse genomic environments (OTM problem). During pre-training, each gene token is masked with a 15% probability and predicted based on its unmasked genome context $\mathcal{G}_M$:

$$\mathcal{L}_{\text{MLM}} = \mathbb{E}_{g \sim \mathcal{G}} \mathbb{E}_M \sum_{i \in M} -\log p(g_i | \mathcal{G}_M), \quad (1)$$

where M denotes the index set of the masked gene.

In addition, considering genetic polymorphism [58, 59], the MGM component focuses on detecting multiple genes that could coexist at a single genomic site, denoted as $\hat{z}_i = [\hat{z}_1, \hat{z}_2, \hat{z}_3, \hat{z}_4]$. This requires the model not only to predict individual genes but also to identify various combinations of genes occurring within the same site. Thus, we enhance the model with a comprehensive loss function, $\mathcal{L}_{\text{MGM}}$, which incorporates feature reconstruction and probability prediction:

$$\mathcal{L}_{\text{MGM}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{y_i^T \hat{z}_i}{\|y_i\| \cdot \|\hat{z}_i\|} \right)^\gamma + \frac{\alpha}{N} \sum_{i=1}^N \|\hat{z}_i - \tilde{y}_i\|_2^2, \quad (2)$$

where γ is a reconstruction loss with the scaled cosine error, and α is a weighting factor to balance the importance of the two loss functions. The first item, feature reconstruction loss (FRL), quantifies the distance between the model prediction $\hat{z}_i$

and its corresponding label y_i . The second item, probability prediction loss (PPL), evaluates the discrepancy between the predicted embedding probability $\hat{z}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$ and the true category probability $\tilde{y}_i = \frac{e^{y_i}}{\sum_{j=1}^C e^{y_j}}$, both processed via the softmax function. C denotes the number of gene combinations and is set to 4.

Triplet metagenomic contrastive framework for many-to-one problem

Contrastive learning

The MGM component enhances the model's ability to learn contextual relationships between genes, helping to alleviate the OTM problem present in metagenomic data. However, when faced with the MTO problem, the model's ability to describe the relationships between sequences with the same function remains weak. For instance, when annotating Enzyme Commission (EC) numbers for partial metagenomic sequences, we observe that sequences sharing the same EC number cluster closely in feature space, while those with distinct EC numbers are more separated [60], with recent advancements [61] optimizing feature extraction for improved clustering. Therefore, we introduce a contrastive learning technique to capture the functional relationships between gene classes, enabling different genes with similar functions to cluster together and further optimize model training. Generally speaking, the objective of contrastive learning is to learn an embedding function $\mathcal{F}$ such that the distance between

Results analysis. FGeneBERT’s performance on CARD-A is significant, as shown in [Table 4](#). This category’s broad range (269 classifications) creates a long-tail distribution, necessitating an understanding of the biological properties of gene sequences for accurate annotation [70]. To mitigate this issue, we adjust the data

Table 4. Macro F1 (% ↑) and Weighted F1 (% ↑) on eight downstream tasks. This includes gene operon prediction on E-K12, ARG prediction on three CARD categories, virulence factors classification on VFDB, enzyme function annotation on ENZYME, microbial pathogens detection on PATRIC, Nitrogen Cycle processes prediction on NCycDB. RF denotes Random Forest, and VT represents Vanilla Transformer. The highest results are highlighted with boldface. The second highest results are highlighted with underline

Method	Operons		ARG prediction						Virus		Enzyme		Pathogen		N-Cycle	
	E-K12		CARD-A		CARD-D		CARD-R		VFDB		ENZYME		PATRIC		NCycDB	
	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1	M.F1	W.F1
RF	20.2	34.8	22.4	35.3	36.1	49.0	47.8	57.6	22.4	38.5	33.6	41.2	25.3	29.8	67.0	71.7
SVM	38.6	45.2	27.6	40.5	33.6	47.2	43.3	66.2	28.0	41.4	31.3	43.6	26.6	31.2	66.9	70.3
KNN	39.9	41.0	36.9	54.4	36.4	51.3	36.2	63.5	27.3	47.1	31.4	42.9	11.0	27.4	68.8	73.2
LSTM	40.4	42.5	47.1	60.3	39.1	62.3	47.5	84.2	36.7	66.3	42.8	51.0	41.3	49.7	71.9	81.2
BiLSTM	38.2	43.8	47.4	61.9	43.5	58.1	58.9	80.3	46.1	72.1	38.7	50.2	43.3	48.5	82.0	88.4
VT	43.3	47.8	57.1	70.0	49.8	68.1	55.7	86.4	58.0	81.0	68.2	75.8	49.8	57.3	84.5	90.7
HyenaDNA	42.4	47.1	50.9	68.2	53.6	78.1	66.2	88.1	<u>61.0</u>	70.4	79.6	83.6	51.1	57.6	92.4	96.0
ESM-2	38.2	42.5	57.2	71.4	56.0	<u>82.1</u>	<u>68.2</u>	90.0	60.7	<u>84.4</u>	<u>92.5</u>	<u>96.7</u>	<u>56.0</u>	<u>67.5</u>	<u>95.8</u>	<u>96.1</u>
NT	45.1	44.8	58.5	72.0	<u>56.2</u>	80.2	68.0	<u>90.3</u>	58.3	71.6	74.1	76.7	46.1	61.9	75.1	86.5
DNABERT2	<u>51.7</u>	<u>52.4</u>	<u>65.2</u>	<u>79.8</u>	51.5	78.7	61.2	88.6	58.2	82.3	85.4	85.2	52.9	60.6	88.6	95.7
Ours	61.8	65.4	78.6	90.1	57.4	85.2	69.4	91.4	75.7	90.2	99.1	98.8	99.3	99.0	99.5	99.2

sampling strategy to increase the frequency of fewer samples, improving the model's prediction accuracy. Table 3 demonstrates FGeneBERT's high prediction accuracy for the CARD-R category, exhibiting superior classification results for both majority and minority classes, with over 85% accuracy. Supplementary Table S8 reveals FGeneBERT has better performance for all 269 AMR Gene Family categories, with 100% classification accuracy for majority categories like CTX, ADC, and CMY, as well as for minor ones like AXC, CRH, and KLUC.

Level 2 Task C: Functional gene prediction → Virulence factors (VF) prediction. This task is to detect microbial elements like bacterial toxins, which enhance pathogen infectivity and exacerbate antimicrobial resistance. Existing methods for metagenomic analysis, particularly those co-predicting ARGs and VFs, are inadequate and suffer from threshold sensitivity issues [71]. We employ the VFDB core dataset [47], a detailed repository of VF characteristics from well-characterized bacterial pathogens, to predict these critical microbial elements. Results analysis. Our model achieves the SOTA results on VFDB, as reported in Table 4. It significantly outperforms the genomic pre-trained model; for instance, M.F1 and W.F1 scores improve by 30% and 9.5%, respectively, compared to DNABERT2. This highlights the limitations of directly applying the genomic pre-trained model to metagenomic data for precise functional annotation. Conversely, ESM-2, as a PLM, excels by leveraging intrinsic protein information in metagenomic data, highlighting its effectiveness.

Level 2 Task D: Functional gene prediction → Enzyme function prediction. This task is critical for understanding metabolism and disease mechanisms in organisms. While traditional methods rely on time-consuming and labor-intensive biochemical experiments, advanced technologies can offer efficient and accurate predictions for large-scale genomic data. We leverage the ENZYME core dataset [48], a comprehensive repository of enzyme nomenclature, to facilitate these predictions. Results analysis. Our experimental results demonstrate FGeneBERT's superior performance on the ENZYME dataset. It outperforms ESM-2, the second-highest method, by approximately 6.62% in M.F1 and 2.09% in W.F1, demonstrating its ability to discern distinct enzyme function characteristics. This observation highlights that our model not only captures gene-protein contextual relationships but also effectively models the relationships between sequences and functions within metagenomic data.

Level 3 Task E: Pathogenicity potential assessment → Genome pathogens prediction. This task assesses the pathogenic potential of pathogens to cope with the public health risks caused by newly emerging pathogens. Accurate deep-learning algorithms are key for the precise identification of pathogens, improving the ability to respond to drug resistance threats. We use PATRIC core dataset [50], which has 5000 pathogenic bacterial sequences across 110 classes. Results analysis. Table 4 shows FGeneBERT's classification of pathogenic bacteria species within PATRIC, demonstrating superior performance over baselines by recognizing crucial genera features. The PATRIC dataset presents a significant challenge due to its large number of categories and sparse data. Baselines generally underperform because they require more data to discern the subtle differences between numerous categories. In contrast, FGeneBERT stands out with M.F1 and W.F1 scores of 99.27% and 99.03%, respectively. This robust performance indicates its advanced learning capability, making it well-suited for high-dimensional classification tasks and highlighting the benefits of using protein-based gene representations for enhanced functional annotation accuracy.

Level 4 Task F: Nitrogen cycle prediction → Nitrogen (N) cycling process prediction. This task focuses on the functional genes related to the N cycle, linking them to environmental and ecological processes. We utilize the NCycDB dataset [51], a comprehensive resource spanning key N cycle gene families and processes, to facilitate this analysis. Results analysis. Table 4 presents FGeneBERT's classification results on NCycDB, suggesting its ability to recognize key features of particular N cycle processes and improve gene family classification by recognizing domains. Although baselines show improved performance on NCycDB compared to PATRIC, due to a larger amount of data per category aiding in discrimination among diverse categories, FGeneBERT still leads with macro F1 (M.F1) and weighted F1 (W.F1) scores of 99.49% and 99.22%, respectively. However, pre-trained baselines require more time and memory for tokenizing large datasets, as analyzed in Model efficiency analysis.

Ablation study

Ablation study on the performance of MGM and TMC

We perform the ablation study to investigate the effectiveness of our proposed components. Table 5 compares the performance of FGeneBERT without MGM and TMC modules across four datasets.

Table 5. Ablation study of M.F1 on CARD-D. *w/o.* MGM denotes FGeneBERT without MGM, *w/o.* Triplet denotes FGeneBERT without TMC

Method	Operons	ARG prediction			
	E-K12	CARD-A	CARD-D	CARD-R	
FGeneBERT	61.8	78.7	57.4	69.4	
<i>w/o.</i> MGM	-8.1	-6.7	-10.5	-6.7	
<i>w/o.</i> Triplet	-7.4	-5.4	-5.8	-3.4	

Bold values highlight the best (maximum) results in each column. Negative values indicate how much lower the corresponding result is compared to the best result.

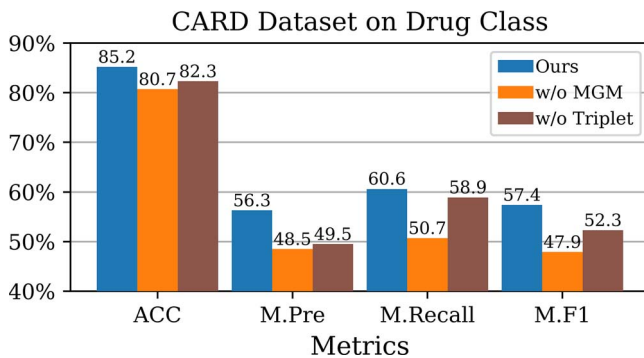


Figure 5. Ablation studies of our proposed modules on four downstream tasks.

Specifically, MGM is designed to address the OTM task, while TMC is tailored to handle the MTO task. The decrease in M.F1 score after removing the MGM and TMC, respectively, highlights their roles in enhancing model performance across different tasks. While TMC contributes to performance, it is evident that MGM has a more substantial impact. Fig. 5 illustrates four more metrics on the CARD dataset.

Ablation study on visualization results of MGM

We conduct a visualization experiment to validate the effectiveness of MGM in the OTM scenario. ATP synthases can exhibit different functions in different organisms to adapt to their respective environmental conditions, even though the basic functions are the same [72].

We collect 1177 ATP synthase sequences from UniProt [73] and color them according to six taxonomies, i.e. Hydrolases (35.6%, dark blue), Isomerases (3.6%, orange), Ligases (35.7%, green), Lyases (10.7%, red), Oxidoreduct (8.9%, light purple) and Transferases (5.3%, brown) (numbers in parentheses indicate category ratios). Figure 6(a) shows the clustering results of ATPase protein embeddings without genome contextual analysis, indicating that more dispersed clustering results of genes in different genome contexts. Figure 6(b) presents the clustering results of ATPase embeddings after our designed protein-based gene embedding, which shows genes belonging to the same category (the same color) apparently cluster together. Concretely, Isomerases (orange) and Transferases (brown), which account for the smallest percentage, cluster together, whereas, in the left plot, these two are scattered. This demonstrates that our proposed MGM can resolve the One-to-Many problem between sequences and functions.

Ablation study on clustering results of TMC

We further explore the impact of integrating TMC into our model on gene operon prediction in the Many-to-One (MTO) scenario.

Table 6. Clustering results of TMC for many-to-one problem

Method	NMI	ARI	Silhouette coefficient
+Tokenizer	0.44	0.34	0.7
+MGM	0.66	0.51	0.72
+TMC	0.72	0.59	0.75

Bold values indicate the best performance in each column.

Initially, we evaluate FGeneBERT without the TMC module. Afterward, we add the TMC module, and K-means are performed using the embeddings from the network's intermediate layers. We compute the normalized mutual information (NMI), adjusted Rand index (ARI), and Silhouette coefficient index to measure the clustering quality. The results in Table 6 show significant improvements in ARI and Silhouette coefficient with TMC, highlighting its effectiveness in enhancing clustering performance across various metrics.

Ablation study on context-aware tokenizer

We replace the FGeneBERT's tokenizer with ESM2 and BPE representations and assessed performance across downstream tasks. Table 7 shows that FGeneBERT outperforms both methods in capturing complex gene sequence-function relationships. Although DNABERT2 compresses sequences effectively with BPE, our context-aware tokenizer demonstrates superior accuracy in metagenomic data analysis. For complete results, see Supplementary Table S6.

Model efficiency analysis

We analyze the time complexity and memory efficiency of our tokenizer compared to four genomic pre-trained methods on six datasets in Fig. 7. Our tokenizer demonstrates superior efficiency, achieving a significant reduction in time and memory usage. Notably, on the NCyc dataset (brown) with 213 501 sequences, ours reduces processing time by 31.05% and memory usage by 94.33% compared to DNABERT2. For the CARD dataset (orange) with 1966 sequences, time, and memory usage decreased by 61.70% and 58.53%. Although HyenaDNA uses less memory on Operons, CARD, VFDB, and ENZYME datasets, it underperforms ours in time cost and overall performance.

To further explore model scalability, we train three FGeneBERT variants—FGeneBERT-T, FGeneBERT-S, and FGeneBERT-B—differing in layers, hidden dimensions, and attention heads. Each variant is trained on 50 million, 100 million, and 150 million sequences to assess the impact on time and memory during pre-training, as shown in Table 8.

Sensitivity analysis

Our sensitivity analysis indicates that FGeneBERT can be optimized effectively using small batch size b without larger performance degradation shown in Fig. 8. This is important for resource-constrained scenarios, highlighting our model maintains good performance even with limited data. We choose a batch size of 1000. Next, we analyze another hyper-parameter of balance ratio α on CARD dataset. FGeneBERT obtains good performance for different values of α , demonstrating its robustness and insensitivity to this hyper-parameter. We set α to be 0.4.

Model configurations

All our experiments are performed on four NVIDIA V100 GPUs and the PyTorch framework. The encoder of FGeneBERT is initialized with Roberta [74]. For pre-training, the parameters are

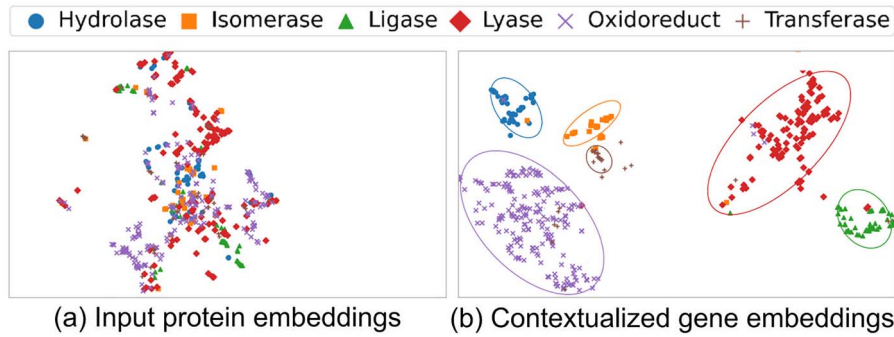


Figure 6. T-SNE visualization of different embeddings for ATP synthases. Each dot denotes a sequence and is grouped according to different functions.

Table 7. The ablation study using protein-based gene representation as a context-aware tokenizer

Dataset	Method	Acc	M.Pre	M.Recall	M.F1	W.Pre	W.Recall	W.F1
E-K12	FGeneBERT	0.68	0.68	0.61	0.61	0.77	0.67	0.65
	ESM2	0.54	0.51	0.48	0.49	0.61	0.51	0.52
	BPE	<u>0.59</u>	<u>0.58</u>	<u>0.55</u>	<u>0.54</u>	<u>0.67</u>	<u>0.54</u>	<u>0.55</u>
CARD-A	FGeneBERT	0.91	0.77	0.8	0.78	0.9	0.91	0.9
	ESM2	0.82	0.74	0.76	0.73	0.87	0.82	0.81
	BPE	<u>0.84</u>	<u>0.75</u>	<u>0.77</u>	<u>0.75</u>	<u>0.88</u>	<u>0.86</u>	<u>0.86</u>

Bold values indicate the best performance, and underlined values indicate the second-best performance.

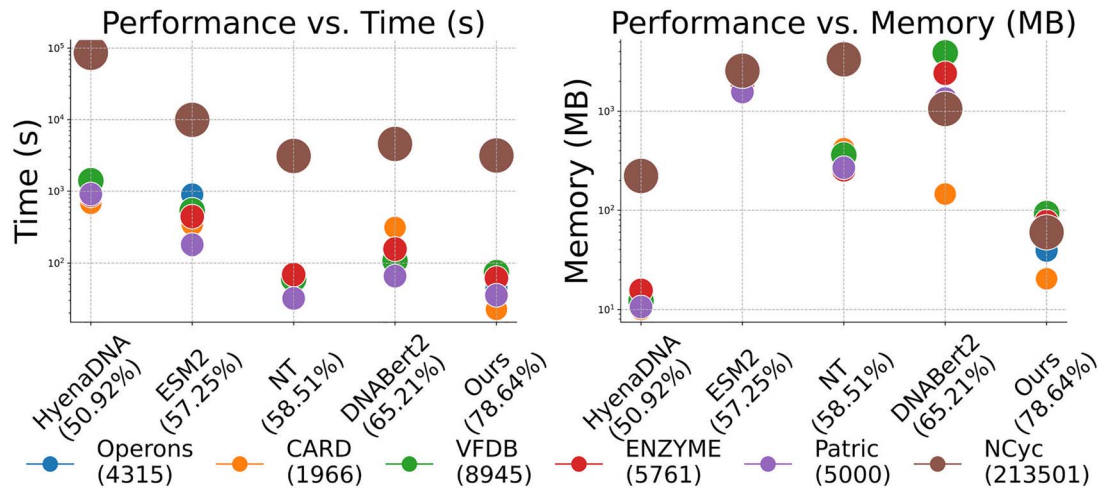


Figure 7. Comparative analysis on tokenization efficiency: time(s) vs. memory (MB). Each point denotes a specific dataset, with the size indicating its scale.

Table 8. Time complexity and memory efficiency of pre-training and scalability of FGeneBERT

Model	P ^a	# L ^b	H ^a	# H ^b	Operons (500/1000/1500 ep)	CARD-A (500/1000/1500 ep)	CARD-D (500/1000/1500 ep)	CARD-R (500/1000/1500 ep)
FGeneBERT-T	50M	10	640	5	50.9 / 61.5 / 63.5	74.6 / 88.6 / 90.7	72.4 / 83.7 / 86.4	77.3 / 90.1 / 91.1
FGeneBERT-S	100M	19	1280	10	55.1 / 65.4 / 65.9	80.4 / 90.1 / 91.2	76.9 / 85.2 / 87.5	81.4 / 91.4 / 93.0
FGeneBERT-B	150M	25	2560	25	57.1 / 66.7 / 67.6	82.7 / 91.1 / 93.2	81 / 87.6 / 90.0	83.7 / 93.4 / 94.8

^aP and H represent the pre-training data and hidden dimension, respectively. ^b# L, and # H represent the number of Layers and Heads, respectively.

set as follows: batch size of 1000, 19 encoder layers, 10 attention heads, an embedding dimension of 1280, and relative position encoding. During the pre-training stage, the model is trained for 500 epochs using the AdamW optimizer [75] with a weight decay of 0.02. The learning rate starts at 1e-5, warms up to 1e-4 over the first 5000 steps, and then decreases back to 1e-5 following a cosine decay schedule. The complete model consists of 954.73

million parameters and requires 2.55 billion FLOPs, as detailed in [Supplementary Table S4](#).

Conclusion

In this paper, we propose a new idea of protein-based gene representation, preserving essential biological characteristics within

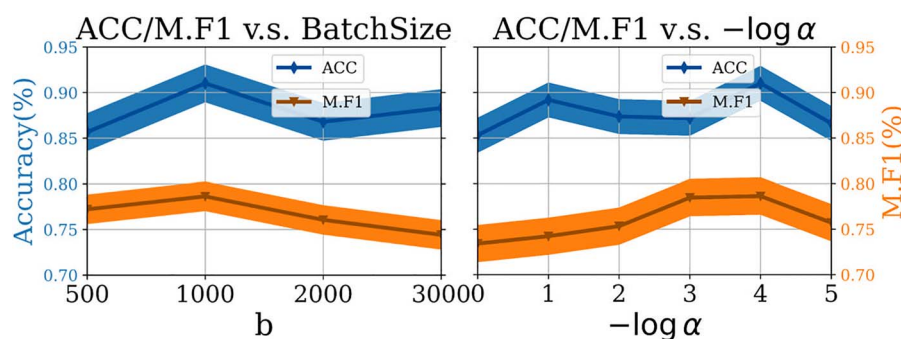


Figure 8. Sensitivity w.r.t. hyper-parameters α , b of CARD dataset on AMR gene family.

each gene sequence. With the new context-aware tokenizer, we propose MGM, a gene group-level pre-training task designed to learn the interactions between genes. Additionally, we develop TMC, a contrastive learning module to generate multiple positive and negative samples to distinguish the gene sequences. MGM and TMC constitute a joint pre-training model, FGeneBERT for metagenomic data. Our experiments and visualizations demonstrate the superior performance of our model. For the future, it remains to be explored how to incorporate multi-omics data, such as metabolomics, into our metagenomic pre-trained model.

Key Points

- Metagenomic analysis faces three key challenges: the lack of context-aware representations, limited capture of inter-gene relationships, and inefficient modeling of gene-function associations in long sequences.
- We introduce FGeneBERT, the first metagenomic pre-trained model that encodes biologically meaningful, context-aware, and function-relevant representations of metagenomic data.
- FGeneBERT employs protein-based gene representations for efficient tokenization of long sequences and uses MGM and TMC modules to model inter-gene relationships and gene-function associations, respectively.
- Comprehensive experiments show that FGeneBERT consistently achieves SOTA performance in metagenomic analysis.

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

Conflict of interest: No competing interest is declared.

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Data availability

All data processed in this study were obtained exclusively from public sources and are available at <https://github.com/jane-pyc/OpenMeta/datasets>. Source codes are freely available at <https://github.com/jane-pyc/OpenMeta/models/FGBERT>.

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